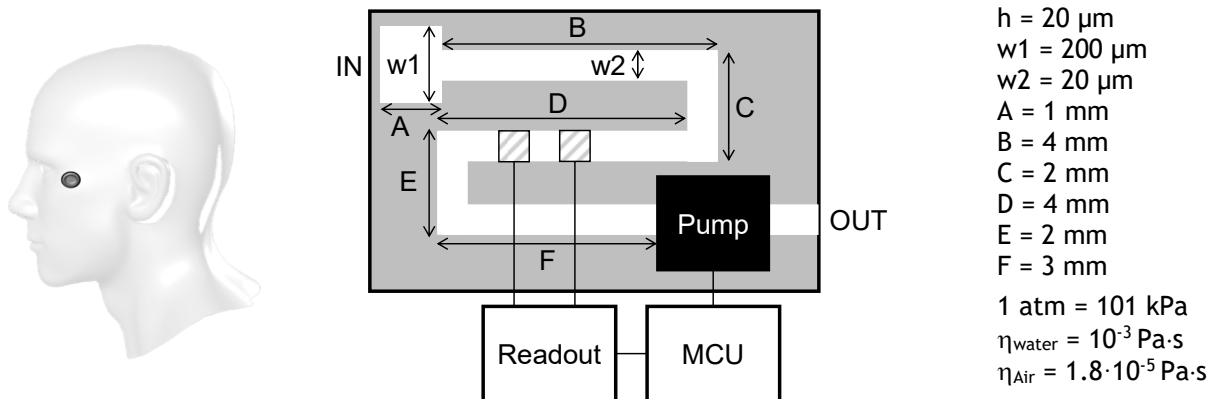


Politecnico di Milano
Biochip A.Y. 2023/2024 - M. Carminati
 Exam of 11.06.2024

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

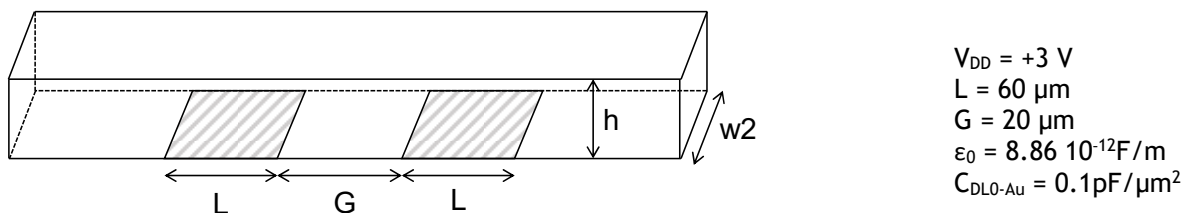
Consider the following wearable microfluidic device for teardrops analysis. It consists of a flexible patch (made of PDMS channels on top of a flexible PCB with electrodes on a Kapton substrate). The system is operated in **aspiration** by an on-chip piezoelectric pump, so that **single tear droplets** are captured into the device from the eye proximity, for real-time electrochemical analysis. When the droplet is electrically detected on the electrodes the pump is stopped and analysis performed. The channel height h is constant.



- Draw the **equivalent model** of the fluidic circuit (including the pump) and compute its parameters in two cases: channel fully filled with water and fully filled with air.
- In the case of water filling, assuming a **flow rate** set by the pump of $2 \mu\text{L}/\text{min}$, find the pressure in all key points and the water speed in all sections. How is speed distributed along the cross-section of the channel?
- What is the impact on the model of point (a) of a small **air bubble** in the water filling case? Where is it more likely that bubbles will get stuck?
- What is the impact of **temperature** variation on the operation of the device?
- Describe the structure and role of **Self Assembled Monolayers** (for molecular detection on chip).
- Draw the device **cross-section** and stack of materials in the region of the electrodes and discuss the impact of the choice of the **metal** for the coplanar sensing electrodes: compare copper vs. gold.
- Propose an alternative technique to handle droplets on chip not requiring pumps.

Exercise 2

Now focus on the sensing part, in particular on the electronic readout the of the coplanar electrodes to measure the **solution resistance** (the tear can be approximated by PBS). The circuit is powered by a **3V battery**.



- Assuming gold plating for the electrodes, draw the equivalent impedance in PBS and compute the known parameter. If the fabrication tolerance produces one electrode 10% longer ($L \cdot 1.1$) and the other electrode 10% shorter ($L \cdot 0.9$), what is the impact of the model values?
- Assuming an ionic solution resistance of **10 kΩ** for the teardrop fully wetting the electrodes, draw the Bode plot of the equivalent impedance spectrum and highlight the frequency region for solution resistance sensing.
- Draw the schematic of the impedance readout circuit and size its parameters (including the bandwidth of the *lock-in* section to allow a detection time below 1 ms).
- Considering only the noise of the solution resistance, size the TIA bandwidth to achieve a noise floor of $1 \text{ nA}_{\text{rms}}$.
- Discuss the impact of replacing the sinusoidal stimulation with a **square wave** of the same period.
- Draw an alternative *ratiometric* configuration for the measurement of solution resistance and compare it with the circuit used here.
- Discuss pros and cons of a microelectronic **CMOS implementation** of the readout circuit.

Multichance students with 30% reduction of the exercises please skip points 1d), 1g), 2f), 2g).